Master's Thesis

Online Imputation of Missing Data in Financial Time Series

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Abstract

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Chapter 1

Introduction

1.1 Motivation

Financial time series data is never fully observed. Limit-order book feeds drop quotes during bursts of order cancellations. Equity markets suspend trading when prices move too violently. Over-the-counter bond markets leave entire yield-curve tenors unquoted for days. Implied volatility surfaces contain large regions where no liquid option trade has occurred. Macro and alternative data arrive at mixed, irregular frequencies, with publication lags that create structural gaps at the daily level. In every one of these settings a practitioner or a researcher faces the same decision: what value should be attributed to the missing observation, and how much uncertainty should be attached to that attribution?

This question is central for market makers and risk managers. For instance, an emerging markets trader could be asked to make a market for an highy illiquid instrument that has not traded for several hours or days. Depending on the instrument, several approaches are possible: for a bond, she could interpolate the missing yield from the rest of the curve; for a derivative, she could use her own pricing model and hope that the imputs are not too stale.[1] The simplest solution is to carry forward the last observed value, but this can lead to systematic mispricing, underestimation of risk, and distortion of correlations among assets.

We aim to take a different path, one that looks for solutions that minimize a loss between the imputed value and the true value, and that preserve as much as possible the statistical properties of the series. In classical machine learning approach, we want to focus on quality of inference rather than interpretability of the model.

Furthermore, and again motivated by practical use cases, we insist on a *causal*

[1] Note that here she is faced by the same problem, i.e. factors (as α, β, ρ in SABR) that may have not been observed in a some time.

imputation: the imputed value at time t must be based only on information available up to time t, and not on any future observation. A market maker does not have the luxury of looking into a magical orb to see what the price will be in the next few seconds. We formalise this constraint through the language of stochastic filtrations, survey and taxonomise the algorithms, from classical statistical methods to modern deep generative models, and evaluate them empirically on representative financial datasets.

1.2 The Causal Constraint

Let us quickly formalise basic notation and the notion of causal imputation.

Following Zhengping Che and Liu (2016), we denote a multivariate time series with D variables of length T as $\mathbf{X}_T = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T)^\top \in \mathbb{R}^{T\times D}$, where $\mathbf{x}_t \in \mathbb{R}^D$ represents the t-th observations (a.k.a., measurements) of all variables and x_t^d denotes the measurement of d-th variable of $\mathbf{x}_t$. Let $s_t \in \mathbb{R}$ denote the time-stamp when the t-th observation is obtained and we assume that the first observation is made at time $t = 0$ ($s_1 = 0$).

Let $(\Omega, \mathcal{G}, \mathbb{P})$ be a probability space and let $\mathbb{F} = \{\mathcal{F}_t\}_{t\geq 1}$ be a filtration representing the information available to a market participant through time. An imputation $\hat{\mathbf{x}}_t^d$ of a missing value $\mathbf{x}_t^d$ is causal if and only if it is $\mathcal{F}_t$-measurable: it may depend on the history $\{\mathbf{X}_s, \mathbf{M}_s\}_{s\leq t}$ of observed values and missingness indicators up to time t, but on nothing later.

The theoretical benchmark for any causal imputation is the conditional expectation

$$\hat{\mathbf{x}}_t^* = \mathbb{E}[\mathbf{x}_t \mid \mathcal{F}_t]\,, \tag{1.1}$$

which is the unique $\mathcal{F}_t$-measurable estimator minimising mean squared error. In the linear Gaussian state-space model this optimum is attained exactly by the Kalman filter prediction step. For the non-linear, non-Gaussian models that may better describe financial data, it can be approximated by recurrent neural networks, diffusion-based generative models, or transformer architectures.

Note the crucial importance of $\sigma(\hat{\mathbf{x}}_t)$ ddd

1.3 Use Cases and Datasets

We organise the empirical investigation around six use cases, each representing a distinct type of financial time series and a distinct pattern of missingness.

UC 1 — Tick and high-frequency data (LOBSTER / TAQ). In a limit-order book, quotes are removed when orders are cancelled or modified. The resulting gaps in the bid–ask spread and mid-price are endogenous to price movement: cancellations cluster around volatile episodes, making the mechanism missing not at random (MNAR). The causal constraint is most binding here, since even a sub-second lookahead constitutes inadmissible information for an execution algorithm. We use the LOBSTER dataset , which provides full ten-level limit-order book snapshots for NASDAQ-listed equities under an academic licence, and the Trade and Quote (TAQ) database via WRDS for cross-validation.

UC 2 — Daily equity prices (CRSP / Yahoo Finance). Trading halts triggered by circuit breakers and regulatory suspensions produce gaps of one to five days in the equity price record. Survivorship bias further distorts panels from which delisted stocks have been removed. The missingness is conditionally missing at random (MAR) during ordinary halts but shifts to MNAR when the halt is caused by undisclosed information. We use the Center for Research in Security Prices (CRSP) daily stock file via WRDS as the primary dataset, with Yahoo Finance (`yfinance` API) as a freely accessible alternative.

UC 3 — Fixed income and yield curves (FRED / ECB SDW). Over-the-counter bond markets produce sparse and irregular quote streams. Off-the-run Treasuries and corporate bonds can go unquoted for days, leaving holes in the yield curve that must be filled causally for real-time risk systems. The missingness pattern ranges from MCAR for the most liquid on-the-run maturities to MNAR for illiquid off-the-run tenors, where the absence of a quote is itself informative about the prevailing liquidity regime. We use the FRED daily yield-curve series (US Treasuries, freely available via the FRED API) and the ECB Statistical Data Warehouse (euro-area sovereign rates) as primary sources.

UC 4 — Macro and mixed-frequency data (FRED-MD). Macro series such as GDP, industrial production, and PMI are published monthly or quarterly, creating structural gaps at the daily frequency required by nowcasting and trading-signal models. Sensor failures and API outages additionally introduce irregular MCAR gaps. The FRED-MD dataset , comprising 128 monthly US macroeconomic series, is widely used as a benchmark in the nowcasting literature, enabling direct comparison with published results.

UC 5 — FX and cryptocurrency (Binance / Dukascopy). Foreign-exchange and cryptocurrency markets operate around the clock, yet liquidity is thin during certain hours or following market stress. The resulting OHLCV gaps are MNAR: illiquidity is endogenous to the bid–ask spread and realised volatility. We use minute-level OHLCV data from the Binance public REST API (cryptocurrency) and tick-level foreign-exchange data from the Dukascopy free historical data service, both of which can be downloaded programmatically without institutional access.

UC 6 — Implied volatility surfaces (CBOE / OptionMetrics). Far out-of-the-money and long-dated options are illiquid; many (strike, maturity) nodes in the implied volatility surface go unquoted on any given day. Imputation here is a two-dimensional problem (strike and maturity) with the additional constraint that any completed surface must respect no-arbitrage conditions: put–call parity and the convexity of option prices as a function of strike. We use the CBOE VIX term-structure series as a freely available proxy and the OptionMetrics IvyDB database via WRDS for the full surface.

The six use cases span all three missingness mechanisms, gap lengths from one tick to several weeks, and data frequencies from millisecond to monthly. For tractability, the main empirical chapters focus on UC 2 (daily equities), UC 3 (yield curves), and UC 5 (high-frequency FX/crypto), which collectively cover the most practically important settings and the widest methodological range. UC 1 and UC 6 are treated as extensions in the final chapter.

1.4 Objectives and Contributions

The thesis makes the following contributions.

1. **A unified causal framework.** We provide a formal definition of causal imputation under a stochastic filtration $\mathbb{F}$, prove the optimality of the conditional expectation $\mathbb{E}[X_t \mid \mathcal{F}_t]$, and introduce the *cost of causality* Δ_t as the MSE gap between the causal optimum and the acausal smoother. This quantity is estimated empirically for each algorithm and dataset, providing a principled measure of how much predictive information resides in future observations.

2. **A taxonomy of causal imputation algorithms.** We survey methods from naïve baselines (LOCF) through classical statistical models (ARIMA, exponential smoothing), state-space filters (Kalman, EKF, particle filter), multiple imputation (EM, MICE), classical machine learning (kNN, missForest,

Gaussian processes, low-rank completion), and deep learning (GRU-D , BRITS Neural ODEs SAITS GP-VAE CSDI For each method we identify whether its standard form is causal, and where it is not, we derive or propose a causal variant.

3. **A standardised benchmark under the causal constraint.** We define a rigorous evaluation protocol with a strict causality assumption in evaluation (no algorithm may access Y_s with $s > t$ at test time), standardised missingness injection under MCAR, MAR, and MNAR at missing rates $\rho \in \{0.1, 0.3, 0.5\}$, and a common suite of metrics including RMSE, MAE, MRE, CRPS, and calibration coverage.

4. **Empirical results on financial datasets.** We report skill scores relative to the LOCF baseline for all algorithms across UC 2, UC 3, and UC 5, with ablations by gap length, missing rate, and missingness mechanism. Our results quantify the practical cost of the causal constraint and identify which algorithm classes offer the best trade-off between statistical performance and computational feasibility for real-time deployment.

1.5 Thesis Structure

The remainder of the thesis is organised as follows.

?? reviews the missing-data and time-series imputation literature, situating our work relative to foundational contributions in statistics

?? develops the mathematical framework: probability-space foundations, Rubin's missingness taxonomy, the formal definition of causal imputation, loss functions and evaluation criteria, the state-space model and Kalman filter as the theoretical gold standard, and extensions to non-linear and non-Gaussian settings.

?? presents the full taxonomy of algorithms, organised from naïve baselines to deep generative models, with a formal causal audit of each method's inference procedure.

?? describes the datasets corresponding to UC 2, UC 3, and UC 5, the missingness-injection methodology, and the train/validation/test splitting protocol that enforces the causal constraint throughout evaluation.

?? reports and discusses the empirical results, including main comparisons, ablations, and an analysis of the estimated cost of causality Δ_t across settings.

?? concludes with a summary of findings, practical recommendations for practitioners, and directions for future research, including the extension to UC 1 (limit-order book) and UC 6 (implied volatility surface) and the open problem of causal diffusion-based imputation under $\mathcal{F}_t$.

Chapter 2

Mathematical Foundations

> "We do not observe what is missing; we only observe that something is missing."
>
> — paraphrasing Little and Rubin (2019)

This chapter develops the mathematical vocabulary used throughout the thesis. Section 2.1 introduces the measure-theoretic framework for stochastic processes. Section 2.2 specialises it to financial time series and records the stylized facts that inform algorithm design. Section 2.3 introduces the formal missing-data model. Section 2.4 presents Rubin's taxonomy adapted to the time-series setting, discusses ignorability, and gives financial interpretations of each mechanism. Section 2.5 formalises the causal imputation constraint and the notion of look-ahead bias. Section 2.6 states the imputation problem precisely and identifies the oracle solution. The chapter closes in Sections 2.7 and 2.8 with identifiability results and the evaluation framework used in subsequent chapters.

2.1 Probability Spaces and Filtrations

Definition 2.1 (Probability space)**.** A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where:

(i) Ω is a non-empty set (the *sample space*);

(ii) $\mathcal{F}$ is a σ-algebra on Ω: a collection of *events* closed under complementation and countable unions, with $\Omega \in \mathcal{F}$;

(iii) $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a *probability measure*: countably additive with $\mathbb{P}(\Omega) = 1$.

The space is *complete* if every subset of a $\mathbb{P}$-null set belongs to $\mathcal{F}$.

Throughout this thesis we work on a fixed complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition 2.2 (Filtration and usual conditions)**.** Let $\mathbb{T}$ be a linearly ordered index set. A *filtration* on $(\Omega, \mathcal{F}, \mathbb{P})$ is an increasing family $\mathbb{F} = (\mathcal{F}_t)_{t \in \mathbb{T}}$ of sub-σ-algebras of $\mathcal{F}$:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F} \qquad \text{for all } s \leq t.$$

The filtration satisfies the *usual conditions* if:

(i) *Completeness*: $\mathcal{F}_0$ contains all $\mathbb{P}$-null sets of $\mathcal{F}$;

(ii) *Right-continuity*: $\mathcal{F}_t = \mathcal{F}_{t+} := \bigcap_{u>t} \mathcal{F}_u$ for all $t \in \mathbb{T}$.

The quadruple $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ is a *filtered probability space.*

Remark 2.3. In discrete time $\mathbb{T} = \{0, 1, 2, \dots\}$ (or a finite horizon $\{0, 1, \dots, T\}$) right-continuity holds automatically, so only completeness needs to be imposed. This is the setting of the present thesis.

The filtration $(\mathcal{F}_t)$ models the *accumulation of information*: $\mathcal{F}_t$ encodes everything that is knowable at time t. It is the central object in the formalisation of the causal constraint.

Definition 2.4 (Adapted and predictable processes)**.** Let $(E, \mathcal{E})$ be a measurable space. A stochastic process $(X_t)_{t \in \mathbb{T}}$ with values in E is:

(i) $\mathbb{F}$-*adapted* if X_t is $\mathcal{F}_t$-measurable for every $t \in \mathbb{T}$;

(ii) $\mathbb{F}$-*predictable* (in discrete time) if X_t is $\mathcal{F}_{t-1}$-measurable for every $t \geq 1$.

An adapted process can depend on all information available *at* time t; a predictable process is determined by information *strictly before* t. Trading strategies, for instance, must be predictable to be implementable.

Definition 2.5 (Natural filtration)**.** The *natural filtration* of a process $(X_t)_{t \in \mathbb{T}}$ is

$$\mathcal{F}_t^X := \sigma(X_s : s \leq t), \qquad t \in \mathbb{T},$$

the smallest filtration to which (X_t) is adapted.

Definition 2.6 (Conditional independence)**.** Random variables X and Y are *conditionally independent given* Z, written $X \perp\!\!\!\perp Y \mid Z$, if

$$\mathbb{P}(X \in A,\, Y \in B \mid Z) = \mathbb{P}(X \in A \mid Z)\, \mathbb{P}(Y \in B \mid Z) \qquad \mathbb{P}\text{-a.s.}$$

for all measurable sets A, B.

2.2 Financial Time Series

Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space with discrete time index $\mathbb{T} = \{1, 2, \dots, T\}$.

Definition 2.7 (Price process and log-returns)**.** Let $d \in \mathbb{N}$. The *price process* is an $\mathbb{F}$-adapted, $\mathbb{R}^d_{>0}$-valued process $(\mathbf{S}_t)_{t\geq 0}$, where the j-th coordinate $S_{j,t}$ denotes the price of asset j at time t. The *multivariate log-return process* is defined component-wise as

$$Y_{j,t} := \log S_{j,t} - \log S_{j,t-1}, \qquad j = 1, \dots, d, \ t \geq 1,$$

so that $\mathbf{Y}_t := (Y_{1,t}, \dots, Y_{d,t})^\top \in \mathbb{R}^d$.

Log-returns are preferred to simple returns $S_{j,t}/S_{j,t-1} - 1$ because they (i) aggregate additively over time: $\log(S_T/S_0) = \sum_{t=1}^T Y_{j,t}$, and (ii) are defined on all of $\mathbb{R}$, which is more convenient for distributional modelling.

Definition 2.8 (Strict and weak stationarity)**.** A process $(\mathbf{Y}_t)_{t\in\mathbb{T}}$ is:

(i) *Strictly stationary* if the joint distribution of $(\mathbf{Y}_{t_1}, \dots, \mathbf{Y}_{t_k})$ equals that of $(\mathbf{Y}_{t_1+h}, \dots, \mathbf{Y}_{t_k+h})$ for all $k \geq 1$, indices $t_1 < \dots < t_k$, and shifts $h \in \mathbb{Z}$;

(ii) *Weakly stationary* (or *covariance-stationary*) if $\mathbb{E}[\mathbf{Y}_t] = \boldsymbol{\mu}$ for all t and $\mathrm{Cov}(\mathbf{Y}_t, \mathbf{Y}_{t+h}) = \Gamma(h)$ for all t, h, where $\Gamma(h)$ depends only on the lag h.

Remark 2.9 (Stylized facts of asset returns). The seminal study of Cont (2001) identifies the following empirical regularities of log-return processes, which must be accounted for in imputation:

(i) *Heavy tails*: the kurtosis of $Y_{j,t}$ exceeds 3 (leptokurtosis), implying more mass in the tails than under normality;

(ii) *Absence of linear autocorrelation*: $\mathrm{Cov}(Y_{j,t}, Y_{j,t-h}) \approx 0$ for $h \geq 1$, consistent with weak-form market efficiency;

(iii) *Volatility clustering*: squared returns $Y_{j,t}^2$ exhibit significant positive autocorrelation for moderate lags h, indicating persistent changes in conditional variance;

(iv) *Leverage effect*: returns and future volatility are negatively correlated, $\mathrm{Cov}(Y_{j,t}, Y_{j,t+h}^2) < 0$ for $h > 0$;

(v) *Cross-sectional correlation*: for assets in the same market, $\mathrm{Cov}(Y_{j,t}, Y_{k,t}) > 0$ for $j \neq k$, a fact exploited by multivariate imputation methods.

Definition 2.10 (ARMA model)**.** A weakly stationary $\mathbb{R}^d$-valued process $(\mathbf{Y}_t)$ follows a *vector ARMA*(p, q) model if

$$\mathbf{Y}_t = \sum_{i=1}^{p} \mathbf{\Phi}_i \, \mathbf{Y}_{t-i} + \boldsymbol{\varepsilon}_t + \sum_{j=1}^{q} \mathbf{\Theta}_j \, \boldsymbol{\varepsilon}_{t-j},$$

where $\mathbf{\Phi}_i, \mathbf{\Theta}_j \in \mathbb{R}^{d\times d}$ are coefficient matrices and $(\boldsymbol{\varepsilon}_t)$ is white noise: $\mathbb{E}[\boldsymbol{\varepsilon}_t] = \mathbf{0}$ and $\mathrm{Cov}(\boldsymbol{\varepsilon}_t, \boldsymbol{\varepsilon}_s) = \mathbf{\Sigma}\, \mathbf{1}_{\{t=s\}}$.

Definition 2.11 (Linear Gaussian state-space model)**.** A *linear Gaussian state-space model* (LGSSM) is defined by:

$$\mathbf{X}_t = \mathbf{A}\, \mathbf{X}_{t-1} + \mathbf{B}\, \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \overset{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{Q}), \tag{2.1}$$

$$\mathbf{Y}_t = \mathbf{C}\, \mathbf{X}_t + \mathbf{D}\, \boldsymbol{\nu}_t, \quad \boldsymbol{\nu}_t \overset{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_d), \tag{2.2}$$

where $\mathbf{X}_t \in \mathbb{R}^m$ is the latent *state*, $\mathbf{Y}_t \in \mathbb{R}^d$ is the *observation*, and $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are matrices of conforming dimensions. The LGSSM admits a recursive causal estimator—the *Kalman filter*—which is developed in Chapter 4.

2.3 The Missing-Data Model

Definition 2.12 (Complete data, missingness indicator, and observed data)**.** For each time $t \in \mathbb{T}$, let $\mathbf{Y}_t^* = (Y_{1,t}^*, \ldots, Y_{d,t}^*)^\top \in \mathbb{R}^d$ denote the *complete* (possibly unobserved) log-return vector. The *missingness indicator* is

$$\mathbf{R}_t = (R_{1,t}, \ldots, R_{d,t})^\top \in \{0,1\}^d, \qquad R_{j,t} := \mathbf{1}_{\left\{Y_{j,t}^* \text{ is observed}\right\}}.$$

The *observed sub-vector* and *missing sub-vector* are, respectively,

$$\mathbf{Y}_t^{obs}(\mathbf{R}_t) := \left(Y_{j,t}^*\right)_{j:\, R_{j,t}=1} \in \mathbb{R}^{d_t^{obs}}, \qquad \mathbf{Y}_t^{mis}(\mathbf{R}_t) := \left(Y_{j,t}^*\right)_{j:\, R_{j,t}=0} \in \mathbb{R}^{d_t^{mis}},$$

where $d_t^{obs} + d_t^{mis} = d$. When the argument $\mathbf{R}_t$ is clear from context we write $\mathbf{Y}_t^{obs}$ and $\mathbf{Y}_t^{mis}$.

Notation 2.13. We write $\mathbf{Y}_{1:t}^* := (\mathbf{Y}_1^*, \ldots, \mathbf{Y}_t^*)$ and similarly $\mathbf{Y}_{1:t}^{obs}$, $\mathbf{R}_{1:t}$. The *observed history* at time t is

$$\mathcal{H}_t := \left(\mathbf{Y}_{1:t}^{obs},\, \mathbf{R}_{1:t}\right).$$

Definition 2.14 (Observed-data filtration)**.** The *observed-data filtration* is

$$\mathcal{F}_t^{obs} := \sigma\!\left(\mathbf{Y}_1^{obs}, \mathbf{R}_1,\, \mathbf{Y}_2^{obs}, \mathbf{R}_2,\, \ldots,\, \mathbf{Y}_t^{obs}, \mathbf{R}_t\right), \qquad t \geq 1,$$

with $\mathcal{F}_0^{obs} := \{\varnothing, \Omega\}$. This is the natural filtration of the observable process $(\mathbf{Y}_t^{obs}, \mathbf{R}_t)_{t \geq 1}$.

The key point is that $\mathcal{F}_t^{obs}$ is *strictly coarser* than the complete-data filtration $\mathcal{F}_t^* := \sigma(\mathbf{Y}_1^*, \ldots, \mathbf{Y}_t^*, \mathbf{R}_1, \ldots, \mathbf{R}_t)$, since we cannot observe $\mathbf{Y}_t^{mis}$.

Definition 2.15 (Joint distribution and canonical factorisations)**.** The joint distribution of complete data and missingness is characterised by a density $p(\mathbf{Y}_{1:T}^*, \mathbf{R}_{1:T}; \boldsymbol{\theta}, \boldsymbol{\psi})$, where $\boldsymbol{\theta}$ parametrises the data-generating process and $\boldsymbol{\psi}$ parametrises the missingness mechanism. Two canonical factorisations are:

(i) *Selection model*:

$$p(\mathbf{Y}^*, \mathbf{R}; \boldsymbol{\theta}, \boldsymbol{\psi}) = p(\mathbf{Y}^*; \boldsymbol{\theta})\; p(\mathbf{R} \mid \mathbf{Y}^*; \boldsymbol{\psi});$$

(ii) *Pattern-mixture model*:

$$p(\mathbf{Y}^*, \mathbf{R}; \boldsymbol{\theta}, \boldsymbol{\psi}) = p(\mathbf{Y}^* \mid \mathbf{R}; \boldsymbol{\alpha})\; p(\mathbf{R}; \boldsymbol{\xi}),$$

where $(\boldsymbol{\alpha}, \boldsymbol{\xi})$ are functions of $(\boldsymbol{\theta}, \boldsymbol{\psi})$.

The selection model directly separates the data distribution from the mechanism; the pattern-mixture model reveals how the distribution of returns changes across missingness patterns.

Remark 2.16 (Recursive structure in time series)*.* In discrete time the data model factors as

$$p(\mathbf{Y}_{1:T}^*; \boldsymbol{\theta}) = p(\mathbf{Y}_1^*; \boldsymbol{\theta}) \prod_{t=2}^{T} p(\mathbf{Y}_t^* \mid \mathbf{Y}_{1:t-1}^*; \boldsymbol{\theta}),$$

reflecting the adapted structure of the process. The missingness mechanism has an analogous factorisation,

$$p(\mathbf{R}_{1:T} \mid \mathbf{Y}_{1:T}^*; \boldsymbol{\psi}) = \prod_{t=1}^{T} p(\mathbf{R}_t \mid \mathbf{Y}_t^*, \mathcal{H}_{t-1}; \boldsymbol{\psi}),$$

which we exploit in the time-series extension of Rubin's taxonomy.

2.4 Rubin's Taxonomy of Missingness Mechanisms

Rubin (1976) introduced a taxonomy of missing-data mechanisms that has become the standard organising framework of the field. The original definitions were stated for cross-sectional data; we adapt them to the time-series setting, following Little and Rubin (2019) and the stochastic-process perspective of Commenges and Jacqmin-Gadda (2022).

2.4.1 Formal Definitions

Definition 2.17 (Missing Completely At Random (MCAR))**.** The missingness at time t is *Missing Completely At Random* (MCAR) if

$$\mathbf{R}_t \perp\!\!\!\perp \mathbf{Y}_t^* \,\Big|\, \mathcal{H}_{t-1},$$

i.e. conditional on the observed history, the missingness indicator is independent of the current complete data vector:

$$\mathbb{P}\Big(\mathbf{R}_t = \mathbf{r} \mid \mathbf{Y}_t^*,\, \mathcal{H}_{t-1}\Big) = \mathbb{P}\Big(\mathbf{R}_t = \mathbf{r} \mid \mathcal{H}_{t-1}\Big) \qquad \mathbb{P}\text{-a.s.}$$

for all $\mathbf{r} \in \{0,1\}^d$.

Definition 2.18 (Missing At Random (MAR))**.** The missingness at time t is *Missing At Random* (MAR) if

$$\mathbf{R}_t \perp\!\!\!\perp \mathbf{Y}_t^{mis} \,\Big|\, \mathbf{Y}_t^{obs},\, \mathcal{H}_{t-1},$$

i.e. conditional on the observed components of $\mathbf{Y}_t^*$ and the observed history, the missingness indicator is independent of the *missing* components:

$$\mathbb{P}\Big(\mathbf{R}_t = \mathbf{r} \mid \mathbf{Y}_t^*,\, \mathcal{H}_{t-1}\Big) = \mathbb{P}\Big(\mathbf{R}_t = \mathbf{r} \mid \mathbf{Y}_t^{obs}(\mathbf{r}),\, \mathcal{H}_{t-1}\Big) \qquad \mathbb{P}\text{-a.s.}$$

Definition 2.19 (Missing Not At Random (MNAR))**.** The missingness at time t is *Missing Not At Random* (MNAR) if it is neither MCAR nor MAR, i.e. the probability of the missingness pattern depends on the *missing* values themselves even after conditioning on $\mathbf{Y}_t^{obs}$ and $\mathcal{H}_{t-1}$.

Remark 2.20 (Hierarchy and terminology)*.* MCAR is a special case of MAR: set $\mathbf{Y}_t^{obs} = \varnothing$ in Theorem 2.18. The three mechanisms are exhaustive and mutually exclusive by construction.

The naming can be misleading: "Missing At Random" does *not* mean that missingness is random in the colloquial sense; it means that missingness can depend on *observed* quantities but not on *unobserved* ones. The term was introduced with this technical meaning by Rubin (1976).

2.4.2 Financial Interpretations

Example 2.21 (MCAR in financial data)**.** A data vendor introduces transmission errors that independently drop each asset's return at each time step with probability $\pi \in (0, 1)$, regardless of the return's magnitude or any market variable: $\mathbb{P}(R_{j,t} = 0) = \pi$ for all j, t, independently. This is MCAR. While it is the most favourable mechanism for inference, it is rarely realistic: data loss in practice tends to correlate with market conditions.

Example 2.22 (MAR in financial data)**.** An illiquid small-cap stock fails to trade during periods of low aggregate market activity. Specifically, suppose $\mathbb{P}(R_{j,t} = 0 \mid Y^*_{j,t}, V_t) = g(V_t)$ for some function g, where V_t (e.g. the total market trading volume) is observed. The probability of missing depends on V_t but not on the stock's own latent return $Y^*_{j,t}$; this is MAR conditional on V_t. Methods that model the missingness as a function of the observed covariates can recover unbiased estimates in this regime.

Example 2.23 (MNAR in financial data)**.** A bond is not priced by the dealer during periods of extreme credit stress, precisely when its value is most anomalous. The probability of non-reporting is an increasing function of $|Y^*_{j,t}|$: $\mathbb{P}(R_{j,t} = 0 \mid Y^*_{j,t})$ is large when $|Y^*_{j,t}|$ is large. This is MNAR. It is arguably the most common pattern in financial data and also the hardest to handle: the missingness is informative about the very quantity we wish to recover.

2.4.3 Ignorability

A central question is when the missingness mechanism can be *ignored* for inference about the data model $\boldsymbol{\theta}$.

Definition 2.24 (Observed-data likelihood)**.** The *observed-data likelihood* for $\boldsymbol{\theta}$ given $\mathcal{H}_T$ is

$$L(\boldsymbol{\theta}; \mathcal{H}_T) := \int p\Big(\mathbf{Y}^*_{1:T}; \boldsymbol{\theta}\Big)\, p\Big(\mathbf{R}_{1:T} \mid \mathbf{Y}^*_{1:T}; \boldsymbol{\psi}\Big)\, d\mathbf{Y}^{mis}_{1:T},$$

obtained by marginalising the joint likelihood over all unobserved values.

Proposition 2.25 (Ignorability under MAR)**.** *Suppose the mechanism is MAR at every t and that the parameters $\boldsymbol{\theta}$ and $\boldsymbol{\psi}$ are* distinct *in the sense that the parameter space factors as $\boldsymbol{\Theta} \times \boldsymbol{\Psi}$ with no shared components. Then*

$$L(\boldsymbol{\theta}; \mathcal{H}_T) \propto p\Big(\mathbf{Y}^{obs}_{1:T}; \boldsymbol{\theta}\Big),$$

i.e. the observed-data likelihood for $\boldsymbol{\theta}$ equals the marginal probability of the observed *data under the complete-data model. Consequently, maximum-likelihood estimation of $\boldsymbol{\theta}$ from $\mathcal{H}_T$ is consistent and asymptotically efficient without modelling $\boldsymbol{\psi}$.*

Proof. Using the recursive factorisation of Theorem 2.16:

$$L(\boldsymbol{\theta}, \boldsymbol{\psi}; \mathcal{H}_T) = \int p(\mathbf{Y}^*; \boldsymbol{\theta}) \prod_{t=1}^{T} p(\mathbf{R}_t \mid \mathbf{Y}_t^*, \mathcal{H}_{t-1}; \boldsymbol{\psi}) \, d\mathbf{Y}^{mis}.$$

Under MAR, $p(\mathbf{R}_t \mid \mathbf{Y}_t^*, \mathcal{H}_{t-1}; \boldsymbol{\psi}) = p(\mathbf{R}_t \mid \mathbf{Y}_t^{obs}, \mathcal{H}_{t-1}; \boldsymbol{\psi})$, so the mechanism factors out of the integral over $\mathbf{Y}^{mis}$:

$$\begin{aligned} L(\boldsymbol{\theta}, \boldsymbol{\psi}; \mathcal{H}_T) &= \prod_{t=1}^{T} p(\mathbf{R}_t \mid \mathbf{Y}_t^{obs}, \mathcal{H}_{t-1}; \boldsymbol{\psi}) \cdot \int p(\mathbf{Y}^*; \boldsymbol{\theta}) \, d\mathbf{Y}^{mis} \\ &= \underbrace{L(\boldsymbol{\psi}; \mathcal{H}_T)}_{\text{mechanism likelihood}} \cdot \underbrace{p(\mathbf{Y}_{1:T}^{obs}; \boldsymbol{\theta})}_{\text{observed-data model}} . \end{aligned}$$

Under parameter distinctness, maximising the joint likelihood over $(\boldsymbol{\theta}, \boldsymbol{\psi})$ decouples into two separate problems; in particular, the maximiser over $\boldsymbol{\theta}$ equals the maximiser of $p(\mathbf{Y}_{1:T}^{obs}; \boldsymbol{\theta})$. □

Remark 2.26. Under MNAR, $p(\mathbf{R}_t \mid \mathbf{Y}_t^*; \boldsymbol{\psi}) \neq p(\mathbf{R}_t \mid \mathbf{Y}_t^{obs}; \boldsymbol{\psi})$ and the mechanism cannot be factored out. Naive maximum-likelihood estimation based on the observed data alone is generally biased. One must either model $\boldsymbol{\psi}$ explicitly (a selection model approach) or impose structural constraints, such as the use of instruments, to achieve identification.

2.5 Causal Imputation and Look-Ahead Bias

We now formalise the central constraint of this thesis.

Definition 2.27 (Imputation rule)**.** An *imputation rule* is a measurable map that assigns, to each (j, t) with $R_{j,t} = 0$, a real-valued random variable $\hat{Y}_{j,t}$ intended as an estimate of $Y_{j,t}^*$.

Definition 2.28 (Causal (online) imputation)**.** An imputation $\hat{Y}_{j,t}$ is *causal* (equivalently, *online* or $\mathcal{F}_t^{obs}$*-measurable*) if

$$\hat{Y}_{j,t} \text{ is } \mathcal{F}_t^{obs}\text{-measurable} \qquad \text{for every } (j, t) \text{ with } R_{j,t} = 0.$$

That is, $\hat{Y}_{j,t}$ depends only on data observed at or before time t.

Definition 2.29 (Smoothing (non-causal) imputation)**.** An imputation $\hat{Y}_{j,t}$ is a *smoothing* imputation if it is $\mathcal{F}_T^{obs}$-measurable for some $T > t$ but is *not* $\mathcal{F}_t^{obs}$-measurable; it uses information from future observations.

Example 2.30 (Causal vs. smoothing: an inventory)**.** (i) *Last Observation Carried Forward* (LOCF): $\hat{Y}_{j,t} = Y^*_{j,\tau}$ where $\tau = \max\{s < t : R_{j,s} = 1\}$. Causal by construction.

(ii) *Linear interpolation* between the last observation before t and the first after t is non-causal: it requires $Y^*_{j,t'}$ for some $t' > t$.

(iii) *Kalman filter* (one-step predictor/update): uses only $\mathcal{F}_t^{obs}$; causal.

(iv) *Kalman smoother* (Durbin and Koopman, 2012, Ch. 4): uses the full-sample $\mathcal{F}_T^{obs}$; non-causal.

(v) A *bidirectional LSTM* (Cao et al., 2018): processes both past and future; non-causal. Its unidirectional (forward-only) variant is causal.

Definition 2.31 (Look-ahead bias)**.** Let $\hat{Y}^{sm}_{j,t}$ be a smoothing imputation and $\hat{Y}^{cl}_{j,t}$ any causal imputation. For a downstream quantity $f_t(\hat{\mathbf{Y}}_t)$ computed using imputed values at time t (e.g. a portfolio weight, a risk estimate, or a factor loading), the *look-ahead bias* is

$$\mathrm{LAB}_t := \mathbb{E}\Big[f_t(\hat{\mathbf{Y}}^{sm}_t)\Big] - \mathbb{E}\Big[f_t(\hat{\mathbf{Y}}^{cl}_t)\Big].$$

In financial backtesting, look-ahead bias inflates apparent in-sample performance and produces results that cannot be replicated in real time.

Proposition 2.32 (Smoothing dominates causal in-sample; causal is required out--of-sample)**.** *Under standard square-integrability, the smoothing imputation* $\hat{Y}^{sm}_{j,t} = \mathbb{E}[Y^*_{j,t} \mid \mathcal{F}_T^{obs}]$ *achieves lower mean squared error than any causal imputation:*

$$\mathbb{E}\Big[(Y^*_{j,t} - \hat{Y}^{sm}_{j,t})^2\Big] \;\le\; \mathbb{E}\Big[(Y^*_{j,t} - \hat{Y}^{cl}_{j,t})^2\Big].$$

However, in a real-time setting only $\mathcal{F}_t^{obs}$ *is available, so* $\hat{Y}^{sm}_{j,t}$ *is* infeasible *and its in-sample advantage is entirely attributable to look-ahead bias.*

Proof. The smoothing estimator is the L^2-projection of $Y^*_{j,t}$ onto the closed linear subspace of $\mathcal{F}_T^{obs}$-measurable square-integrable random variables. Any causal estimator $\hat{Y}^{cl}_{j,t}$ belongs to the $\mathcal{F}_t^{obs}$-measurable subspace, which is a *subspace* of the former since $\mathcal{F}_t^{obs} \subseteq \mathcal{F}_T^{obs}$. The L^2-projection onto a smaller subspace cannot have smaller residual norm (Pythagoras in Hilbert space), so the inequality follows. □

2.6 Problem Statement

We are now in a position to state the central problem precisely.

Definition 2.33 (Causal imputation problem)**.** Given:

- a complete-data process $(\mathbf{Y}_t^*)_{t\in\mathbb{T}}$,
- an observed-data filtration $(\mathcal{F}_t^{obs})_{t\in\mathbb{T}}$,
- a loss function $\ell : \mathbb{R} \times \mathbb{R} \to \mathbb{R}_{\geq 0}$,

the *causal imputation problem* is: for each (j,t) with $R_{j,t} = 0$, find an $\mathcal{F}_t^{obs}$-measurable random variable $\hat{Y}_{j,t}$ that minimises

$$\mathbb{E}\Big[\ell(\hat{Y}_{j,t},\, Y_{j,t}^*) \mid \mathcal{F}_t^{obs}\Big].$$

Proposition 2.34 (Oracle causal imputation under squared loss)**.** *Under $\ell(a,b) = (a-b)^2$, the unique minimiser of $\mathbb{E}[(Y_{j,t}^* - \hat{Y}_{j,t})^2 \mid \mathcal{F}_t^{obs}]$ over all $\mathcal{F}_t^{obs}$-measurable random variables is the conditional expectation*

$$\hat{Y}_{j,t}^\star = \mathbb{E}\Big[Y_{j,t}^* \mid \mathcal{F}_t^{obs}\Big].$$

Proof. For any $\mathcal{F}_t^{obs}$-measurable $\hat{Y}_{j,t}$, the orthogonality of the L^2-projection gives

$$\mathbb{E}\Big[(\hat{Y}_{j,t} - Y_{j,t}^*)^2 \mid \mathcal{F}_t^{obs}\Big] = \Big(\hat{Y}_{j,t} - \mathbb{E}[Y_{j,t}^* \mid \mathcal{F}_t^{obs}]\Big)^2 + \mathrm{Var}(Y_{j,t}^* \mid \mathcal{F}_t^{obs}).$$

The first term is non-negative and equals zero if and only if $\hat{Y}_{j,t} = \mathbb{E}[Y_{j,t}^* \mid \mathcal{F}_t^{obs}]$. □

Remark 2.35. In practice $\mathbb{E}[Y_{j,t}^* \mid \mathcal{F}_t^{obs}]$ is not available in closed form unless the complete-data model and missingness mechanism are fully specified and analytically tractable. All algorithms studied in Chapter 4 can be understood as different approximations to this oracle estimator within their respective model classes.

2.7 Identifiability

Imputation is meaningful only if the complete-data distribution is, at least partially, identifiable from the observed data.

Definition 2.36 (Identifiability)**.** The parameter $\boldsymbol{\theta}$ is *identifiable* from the observed data if for any $\boldsymbol{\theta} \neq \boldsymbol{\theta}'$,

$$p(\mathbf{Y}_{1:T}^{obs}; \boldsymbol{\theta}) \neq p(\mathbf{Y}_{1:T}^{obs}; \boldsymbol{\theta}')$$

with positive probability.

Proposition 2.37 (Identifiability under MCAR)**.** *Under MCAR, the observed-data marginal satisfies* $p(\mathbf{Y}^{obs};\boldsymbol{\theta}) = \int p(\mathbf{Y}^*;\boldsymbol{\theta})\, d\mathbf{Y}^{mis}$*, and any parametric family* $\{p(\cdot\,;\boldsymbol{\theta})\}$ *that is identifiable in the complete-data setting remains identifiable from the observed data.*

Proposition 2.38 (Generic non-identifiability under MNAR)**.** *Under MNAR, for any observed-data distribution there exist (generically) multiple pairs* $(\boldsymbol{\theta}, \boldsymbol{\psi})$ *with different complete-data distributions* $p(\mathbf{Y}^*;\boldsymbol{\theta})$ *and different mechanisms* $p(\mathbf{R} \mid \mathbf{Y}^*;\boldsymbol{\psi})$ *that are observationally equivalent. The complete-data distribution is not identifiable from observed data alone without additional structural restrictions on* $\boldsymbol{\psi}$*.*

Remark 2.39. Theorem 2.38 has a stark practical consequence: under MNAR, any imputation algorithm is making untestable assumptions about the relationship between missing values and their own absence. Sensitivity analysis with respect to the assumed mechanism $\boldsymbol{\psi}$ is therefore essential whenever MNAR cannot be ruled out, as is often the case in financial data (Theorem 2.23).

2.8 Evaluation Framework

Definition 2.40 (Test missingness set)**.** Given a dataset $\{\mathbf{Y}_t^*\}_{t=1}^T$ with fully known ground truth and an artificially imposed missingness pattern $\{\mathbf{R}_t\}_{t=1}^T$, the *test set* is

$$\mathcal{M}_{test} := \Big\{(j,t) : R_{j,t} = 0\Big\}.$$

Definition 2.41 (Imputation error metrics)**.** For a point imputation $\{\hat{Y}_{j,t}\}_{(j,t)\in\mathcal{M}_{test}}$, define:

$$\text{RMSE} := \sqrt{\frac{1}{|\mathcal{M}_{test}|} \sum_{(j,t)\in\mathcal{M}_{test}} \Big(\hat{Y}_{j,t} - Y_{j,t}^*\Big)^2}, \tag{2.3}$$

$$\text{MAE} := \frac{1}{|\mathcal{M}_{test}|} \sum_{(j,t)\in\mathcal{M}_{test}} \Big|\hat{Y}_{j,t} - Y_{j,t}^*\Big|, \tag{2.4}$$

$$\text{MAPE} := \frac{1}{|\mathcal{M}_{test}|} \sum_{(j,t)\in\mathcal{M}_{test}} \frac{|\hat{Y}_{j,t} - Y_{j,t}^*|}{|Y_{j,t}^*| + \epsilon}, \tag{2.5}$$

where $\epsilon > 0$ is a small stability constant.

Remark 2.42 (Metric choice and heavy tails)*.* Since financial returns are heavy-tailed (Theorem 2.9), RMSE is sensitive to extreme observations that may dominate the average. We therefore report both RMSE and MAE in the empirical chapters. For distributional quality we additionally compute the first-order *Wasserstein distance* $\mathcal{W}_1(\hat{p}, p^*)$ between the empirical distributions of imputed and true values (Choudhury et al., 2021).

Assumption 2.43 (Evaluation protocol)**.** Throughout the empirical study we adopt the following protocol:

(i) Missingness is *artificially* imposed on an otherwise complete dataset, so ground-truth values are available for evaluation.

(ii) All causal methods observe only $\mathcal{H}_t$ when imputing $Y^*_{j,t}$; the evaluation confirms causality by verifying that no index $t' > t$ appears in the computation graph.

(iii) Performance is reported separately for each mechanism (MCAR, MAR, MNAR) and for multiple missingness rates $\rho \in \{5\%, 10\%, 20\%, 40\%\}$.

Chapter 3

Literature Review

Chapter 4

Imputation Methods

4.1 Baseline Methods

4.1.1 Zero Imputation

4.2 Machine-Learning Methods

4.2.1 Causal Transformer

Chapter 5

Results and Discussion

Chapter 6

Conclusion

6.1 Summary

6.2 Limitations

6.3 Future Work

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